

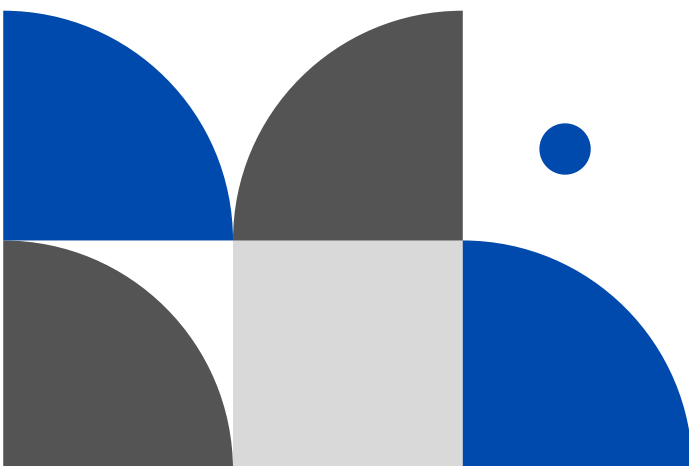
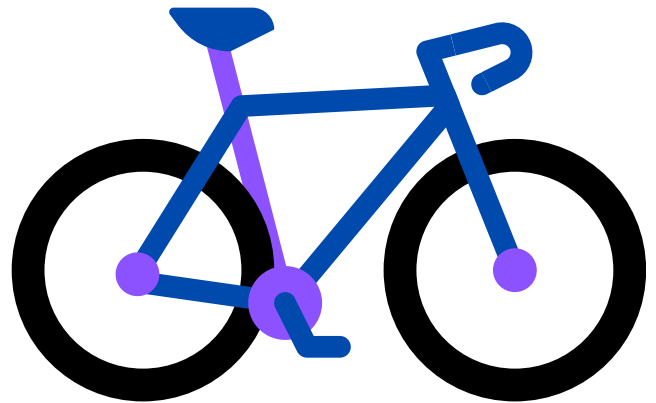
● MINI PROJECT



BEE - CYCLE

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- **Top 10 Produk paling Banyak Dibeli**
- **Produk dengan profit Terbesar**
- **Toko Cabang dengan Jumlah Customer dan Total Transaksi Tertinggi**
- **Voucher Diskon untuk Customer dengan Total Transaksi Tertinggi**
- **Transaksi Terbesar berdasarkan Gender dan Umur**
- **Produk dengan Total Transaksi Terendah per Tahun**
- **Kota dengan Transaksi Terbesar berdasarkan Kategori Produk**
- **Biaya Produk Terbesar pada Tahun 2017-2019**

BACKGROUND DATA

Bee-Cycle

Bee-Cycle adalah perusahaan manufaktur sepeda.

Mereka menjual 4 kategori produk, antara lain:

- Bikes (Touring, Mountain, etc)
- Accessories (Helm, Lights, etc)
- Clothing (Sock, Jersey, etc)
- Component (Wheels, Pedal etc)

Architecture

Secara Arsitektur, Bee-Cycle menggunakan

Database PostgreSQL dan memiliki 5 tabel utama :

- **dim_product** : Informasi produk
- **dim_customer** : Informasi personal customer
- **dim_territory** : Informasi toko cabang
- **dim_geography** : Informasi geografi customer
- **fact_sales** : Informasi detail transaksi

TOOLS



CLOUD DATABASE

dim_geography

```
CREATE TABLE public.dim_geography (  
  geography_id int4 primary key,  
  city varchar(50),  
  state_province_code varchar(5),  
  state_province_name varchar(50),  
  country_region_code varchar(5),  
  english_country_region_name varchar(50),  
  postal_code varchar(50))
```

dim_geography	
123	geography_id
A-Z	city
A-Z	state_province_code
A-Z	state_province_name
A-Z	country_region_code
A-Z	english_country_region_name
A-Z	postal_code

dim_customer

```
CREATE TABLE public.dim_customer (  
  customer_id int4 primary key,  
  geography_id int4,  
  customer_name varchar(100),  
  birthdate date,  
  maritalstatus varchar(5),  
  gender varchar(5),  
  datefirstpurchase date  
  foreign key(geography_id) references dim_geography(geography_id))
```

dim_customer	
123	customer_id
123	geography_id
A-Z	customer_name
🕒	birthdate
A-Z	maritalstatus
A-Z	gender
🕒	datefirstpurchase

dim_product

```
CREATE TABLE public.dim_product (  
  product_id int4 primary key,  
  product_name varchar(50),  
  model_name varchar(50),  
  color varchar(15),  
  size_range varchar(15),  
  "cost" numeric(10),  
  normal_price numeric(10),  
  sub_category varchar(50),  
  category varchar(50))
```

dim_product	
123	product_id
A-Z	product_name
A-Z	model_name
A-Z	color
A-Z	size_range
123	cost
123	normal_price
A-Z	sub_category
A-Z	category

CLOUD DATABASE

dim_territory

```
CREATE TABLE public.dim_territory (  
  territory_id int4 primary key,  
  region varchar(30),  
  country varchar(30),  
  "groups" varchar(30))
```

dim_territory
123 territory_id
A-Z region
A-Z country
A-Z groups

fact_sales

```
CREATE TABLE public.fact_sales (  
  order_detail_id varchar(50) primary key,  
  order_date date,  
  product_id int4,  
  customer_id int4,  
  territory_id int4,  
  sales_order_number varchar(50),  
  sales_order_line_number int4,  
  quantity int4,  
  unitprice_rupiah numeric(12),  
  totalprice_rupiah numeric(12),  
  totalcost_rupiah numeric(12),  
  shippingprice_rupiah numeric(12),  
  foreign key(product_id) references dim_product(product_id),  
  foreign key(customer_id) references dim_customer(customer_id),  
  foreign key(territory_id) references dim_territory(territory_id))
```

fact_sales
A-Z order_detail_id
🕒 order_date
123 product_id
123 customer_id
123 territory_id
A-Z sales_order_number
123 sales_order_line_number
123 quantity
123 unitprice_rupiah
123 totalprice_rupiah
123 totalcost_rupiah
123 shippingprice_rupiah

TABLE OVERVIEW

dim_geography

Column:

1. **geography_id** : Identifier [Primary Key]
2. **city** : Nama kota
3. **state_province_code** : Kode provinsi
4. **state_province_name** : Nama provinsi
5. **country_region_code** : Kode negara
6. **english_country_region_name** : Nama negara
7. **postal_code** : Kode pos

QUERY

```
SELECT *  
FROM dim_geography dg  
LIMIT 5
```

123 geography_id	AZ city	AZ state_province_code	AZ state_province_name	AZ country_region_code	AZ english_country_region_name	AZ postal_code
2	Coffs Harbour	NSW	New South Wales	AU	Australia	2450
3	Darlinghurst	NSW	New South Wales	AU	Australia	2010
4	Goulburn	NSW	New South Wales	AU	Australia	2580
5	Lane Cove	NSW	New South Wales	AU	Australia	1597
6	Lavender Bay	NSW	New South Wales	AU	Australia	2060

dim_customer

Column:

1. **customer_id** : Identifier [**Primary Key**]
2. **customer_name** : Nama customer
3. **birthdate** : Tanggal lahir
4. **maritalstatus** : Status pernikahan
5. **gender** : Jenis kelamin
6. **datefirstpurchase** : Tanggal pertama kali transaksi

QUERY

```
SELECT *  
FROM dim_customer dc  
LIMIT 5
```

123 customer_id	123 geography_id	AZ customer_name	birthdate	AZ maritalstatus	AZ gender	datefirstpurchase
11,000	26	Jon Yang	1986-04-08	M	M	2016-07-22
11,001	37	Eugene Huang	1985-05-14	S	M	2016-07-18
11,002	31	Ruben Torres	1985-08-12	M	M	2016-07-10
11,004	19	Elizabeth Johnson	1988-08-08	S	F	2016-07-26
11,005	22	Julio Ruiz	1985-08-05	S	M	2016-07-02

dim_product

Column:

1. **product_id** : Identifier [**Primary Key**]
2. **product_name** : Nama produk
3. **model_name** : Jenis model
4. **color** : Warna produk
5. **size_range** : Ukuran produk
6. **cost** : Biaya produksi
7. **normal_price** : Harga produk
8. **sub_category** : Sub-kategori produk
9. **category** : Kategori produk

QUERY

```
SELECT *  
FROM dim_product dp  
LIMIT 5
```

123 product_id	AZ product_name	AZ model_name	AZ color	AZ size_range	123 cost	123 normal_price	AZ sub_category	AZ category
210	HL Road Frame - Black, 58	HL Road Frame	Black	54-58 CM	11,000	11,000	Road Frames	Components
211	HL Road Frame - Red, 58	HL Road Frame	Red	54-58 CM	11,000	11,000	Road Frames	Components
480	Patch Kit/8 Patches	Patch kit	NA	NA	11,991	32,060	Tires and Tubes	Accessories
529	Road Tire Tube	Road Tire Tube	NA	NA	20,892	55,860	Tires and Tubes	Accessories
477	Water Bottle - 30 oz.	Water Bottle	NA	NA	26,128	69,860	Bottles and Cages	Accessories

TABLE OVERVIEW

dim_territory

Column:

1. **territory_id** : Identifier [**Primary Key**]
2. **region** : Nama wilayah
3. **country** : Nama negara
4. **groups** : Grup negara

QUERY

```
SELECT *  
FROM dim_territory dt  
LIMIT 5
```

123 territory_id ▾	A-Z region ▾	A-Z country ▾	A-Z groups ▾
1	Northwest	United States	North America
2	Northeast	United States	North America
3	Central	United States	North America
4	Southwest	United States	North America
5	Southeast	United States	North America

fact_sales

Column:

1. **order_detail_id**: Identifier [**Primary Key**]
2. **order_date** : Tanggal pemesanan
3. **product_id** : Identifier produk
4. **customer_id** : Identifier customer
5. **territory_id**: Identifier territory
6. **sales_order_number** : Nomor sales order
7. **sales_order_line_number** : Jumlah line produk pada 1 sales order
8. **quantity** : Jumlah barang yang dipesan
9. **unitprice_rupiah** : Harga barang per 1 produk
10. **totalprice_rupiah** : Harga total (quantity * unitprice)
11. **totalcost_rupiah** : Biaya total per produk
12. **shippingprice_rupiah** : Biaya pengiriman

QUERY

```
SELECT *  
FROM fact_sales fs  
LIMIT 5
```

A-Z order_detail_id ▾	🕒 order_date ▾	123 product_id ▾	123 customer_id ▾	123 territory_id ▾	A-Z sales_order_number ▾
SO43698-1	2016-07-01	346	28,389	7	SO43698
SO43704-1	2016-07-02	351	11,005	9	SO43704
SO43705-1	2016-07-02	344	11,011	9	SO43705
SO43713-1	2016-07-05	310	27,601	4	SO43713
SO43714-1	2016-07-05	311	13,591	10	SO43714

123 sales_order_line_number ▾	123 quantity ▾	123 unitprice_rupiah ▾	123 totalprice_rupiah ▾	123 totalcost_rupiah ▾	123 shippingprice_rupiah ▾
1	1	47,599,900	47,599,900	26,770,200	1,190,000
1	1	47,249,900	47,249,900	26,573,300	1,181,250
1	1	47,599,900	47,599,900	26,770,200	1,190,000
1	1	50,095,800	50,095,800	30,398,100	1,252,400
1	1	50,095,800	50,095,800	30,398,100	1,252,400

ACTIONABLE KNOWLEDGE

1. Top 10 Produk paling Banyak Dibeli

QUERY

```
WITH transaksi AS (  
    SELECT fs.product_id, COUNT(DISTINCT fs.order_detail_id) AS count_ord  
    FROM fact_sales fs  
    GROUP BY product_id  
)  
SELECT dp.product_id, dp.product_name, dp.model_name, dp.category,  
dp.sub_category, t.count_ord  
FROM dim_product dp  
LEFT JOIN transaksi t ON t.product_id = dp.product_id  
WHERE t.count_ord is not null  
ORDER BY t.count_ord DESC  
LIMIT 10;
```

OUTPUT

product_id	product_name	model_name	category	sub_category	count_ord
477	Water Bottle - 30 oz.	Water Bottle	Accessories	Bottles and Cages	363
480	Patch Kit/8 Patches	Patch kit	Accessories	Tires and Tubes	262
478	Mountain Bottle Cage	Mountain Bottle Cage	Accessories	Bottles and Cages	228
222	Sport-100 Helmet, Blue	Sport-100	Accessories	Helmets	221
214	Sport-100 Helmet, Red	Sport-100	Accessories	Helmets	201
528	Mountain Tire Tube	Mountain Tire Tube	Accessories	Tires and Tubes	200
537	HL Mountain Tire	HL Mountain Tire	Accessories	Tires and Tubes	199
479	Road Bottle Cage	Road Bottle Cage	Accessories	Bottles and Cages	191
225	AWC Logo Cap	Cycling Cap	Clothing	Caps	190
485	Fender Set - Mountain	Fender Set - Mountain	Accessories	Fenders	188

CONCLUSION

Dari hasil analisis data penjualan, diketahui bahwa produk dengan kategori **aksesoris** menempati posisi teratas dalam daftar Top 10 produk paling banyak dibeli selama periode analisis.

BUSSINESS RECOMMENDATION

- Optimalkan stok di kategori ini untuk mencegah kehabisan produk saat permintaan tinggi.
- Perkuat strategi promosi dengan menjalankan kampanye bertema “gaya hidup sehat” dan “bersepeda sebagai transportasi ramah lingkungan”

ACTIONABLE KNOWLEDGE

2. Produk dengan profit terbesar

QUERY

```
SELECT
    dp.sub_category,
    dp.category,
    SUM(t.totalprice_rupiah - t.totalcost_rupiah ) AS profit,
    COUNT(t.order_detail_id ) AS jumlah_transaksi
FROM fact_sales t
LEFT JOIN dim_product dp ON t.product_id = dp.product_id
GROUP BY 1,2
ORDER BY profit DESC
LIMIT 10;
```

OUTPUT

AZ sub_category ▼	AZ category ▼	123 profit ▼	123 jumlah_transaksi ▼
Mountain Bikes	Bikes	17,100,316,460	1,113
Road Bikes	Bikes	9,813,036,710	670
Touring Bikes	Bikes	4,858,823,920	467
Helmets	Accessories	183,071,244	597
Tires and Tubes	Accessories	161,341,388	1,254
Bottles and Cages	Accessories	50,885,080	782
Hydration Packs	Accessories	44,819,676	93
Jerseys	Clothing	41,678,216	247
Fenders	Accessories	36,215,004	188
Bike Racks	Accessories	30,498,720	29

CONCLUTION

Kategori **Bikes** dan **Accessories** menempati 10 besar penyumbang profit di perusahaan. Namun untuk sub-kategori **Mountain Bikes** (sepeda gunung) mempunyai profit yg terbesar dan berbanding lurus dengan banyaknya transaksi. Ini menunjukkan minat customer yang tinggi terhadap kategori tersebut

BUSSINESS RECOMMENDATION

1. Meningkatkan penjualan kategori “Mountain Bikes” di semua cabang toko, supaya pendapatan maksimal.
2. Berikan kupon atau voucher setiap pembelian produk tersebut.

ACTIONABLE KNOWLEDGE

3. Toko Cabang dengan Jumlah Customer dan Total Transaksi Tertinggi

QUERY

```
SELECT
    dt.region,
    dt.country,
    SUM(t.totalprice_rupiah) AS total_transaksi,
    COUNT(DISTINCT t.customer_id) AS jumlah_customer
FROM fact_sales t
LEFT JOIN dim_territory dt ON t.territory_id = dt.territory_id
GROUP BY 1,2
ORDER BY 3 DESC
```

OUTPUT

A-Z region ▼	A-Z country ▼	123 total_transaksi ▼	123 jumlah_customer ▼
Australia	Australia	27,054,914,900	357
Southwest	United States	10,311,467,820	243
United Kingdom	United Kingdom	10,178,199,560	135
Germany	Germany	9,158,806,020	121
France	France	9,083,334,580	140
Northwest	United States	6,780,945,220	175
Canada	Canada	4,624,898,160	184
Southeast	United States	33,610,680	2

CONCLUTION

Berdasarkan hasil analisis, dapat kita lihat bahwa untuk Toko Cabang di **Australia** memiliki **total transaksi** dan **jumlah customer** paling tinggi dibandingkan cabang lain, sehingga potensinya sangat tinggi untuk dikembangkan. Selain itu, Toko Cabang di Southeast US merupakan toko dengan performa terendah, maka perlunya dilakukan evaluasi.

BUSSINESS RECOMMENDATION

1. Meningkatkan kapasitas, variasi produk, serta strategi pemasaran karena memiliki potensi pertumbuhan tinggi di Australia.
2. Di Souteast US, telusuri penyebab rendahnya performa dan lakukan perbaikan strategi penjualan serta promosi.

ACTIONABLE KNOWLEDGE

4. Voucher Diskon untuk Customer dengan Total Transaksi Tertinggi

QUERY

```
SELECT
    dc.customer_name,
    dg.city,
    COUNT(t.sales_order_number) AS total_transaksi
FROM fact_sales t
LEFT JOIN dim_customer dc ON t.customer_id = dc.customer_id
LEFT JOIN dim_geography dg ON dc.geography_id = dg.geography_id
GROUP BY dc.customer_id, 2
ORDER BY 3 DESC
LIMIT 10
```

OUTPUT

A-Z customer_name ▼	A-Z city ▼	123 total_transaksi ▼
Samantha Jenkins	Shawnee	43
Henry Garcia	Cliffside	41
Dalton Perez	Haney	38
Charles Jackson	Royal Oak	38
April Shan	Cergy	35
Mason Roberts	Haney	35
Hailey Patterson	Langley	34
Jason Griffin	Victoria	34
Ryan Thompson	Oak Bay	34
Fernando Barnes	N. Vancouver	33

CONCLUTION

Dapat dilihat di tabel yang menunjukkan 10 besar customer dengan total transaksi tertinggi. Para customer ini dapat diberikan voucher diskon supaya lebih rajin lagi berbelanja.

BUSSINESS RECOMMENDATION

1. Memberikan voucher belanja untuk customer.
2. Membuat sistem poin atau membership agar customer merasa dihargai dan termotivasi untuk terus bertransaksi di toko.

ACTIONABLE KNOWLEDGE

5. Transaksi Terbesar berdasarkan Gender dan Umur

QUERY

```
WITH customer_age AS (  
    SELECT dc.customer_id, dc.gender, dc.customer_name,  
           (date_part('year',CURRENT_DATE) - date_part('year',dc.birthdate)) AS age  
    FROM dim_customer dc  
) , ageGroup AS (  
    SELECT (CASE WHEN ca.age <= 20 THEN 'Group <= 20'  
                WHEN ca.age BETWEEN 21 AND 40 THEN 'Group 21-40'  
                WHEN ca.age BETWEEN 41 AND 60 THEN 'Group 41-60'  
                ELSE 'Group > 60'  
            END) AS age_group, ca.gender, ca.customer_id, t.order_detail_id  
    FROM fact_sales t  
    LEFT JOIN customer_age ca ON t.customer_id = ca.customer_id  
) SELECT ag.gender, ag.age_group,  
COUNT(ag.order_detail_id) AS total_transaksi  
FROM ageGroup ag  
WHERE ag.gender IS NOT NULL  
GROUP BY 1,2  
ORDER BY 3 DESC
```

OUTPUT

AZ gender	AZ age_group	123 total_transaksi
F	Group 41-60	1,593
M	Group 41-60	1,477
F	Group 21-40	1,307
M	Group 21-40	1,150
M	Group > 60	239
F	Group > 60	187

CONCLUTION

Gender F atau Perempuan / Wanita dengan rentang usia 41 - 60 tahun mempunyai total transaksi terbesar di semua toko, dengan total transaksi di atas 1500. Diikuti dengan gender M atau Laki-laki dengan total transaksi 1477.

BUSSINESS RECOMMENDATION

1. Buat strategi pemasaran khusus, seperti promo produk yang relevan dengan kebutuhan mereka atau program loyalitas eksklusif untuk mempertahankan pelanggan utama.

ACTIONABLE KNOWLEDGE

6. Produk dengan Total Transaksi Terendah per Tahun

QUERY

```
WITH ranking AS (  
    SELECT dp.sub_category, dp.category,  
           (date_part('year',t.order_date )) AS tahun,  
           COUNT(t.sales_order_number) AS jumlah_transaksi,  
           ROW_NUMBER() OVER (PARTITION BY (date_part('year',t.order_date )) ORDER  
BY COUNT(t.sales_order_line_number)) AS rank  
    FROM fact_sales t  
    LEFT JOIN dim_product dp ON t.product_id = dp.product_id  
    GROUP BY 1, 2, 3 )  
SELECT  
    r.tahun, r.sub_category, r.category, r.jumlah_transaksi  
FROM ranking r  
WHERE r."rank" <= 3  
ORDER BY 1, 4 , r."rank"
```

OUTPUT

123 tahun	AZ sub_category	AZ category	123 jumlah_transaksi
2,016	Mountain Bikes	Bikes	121
2,016	Road Bikes	Bikes	218
2,017	Mountain Bikes	Bikes	271
2,017	Road Bikes	Bikes	290
2,018	Shorts	Clothing	8
2,018	Bike Stands	Accessories	13
2,018	Vests	Clothing	19
2,019	Bike Racks	Accessories	6
2,019	Bike Stands	Accessories	7
2,019	Shorts	Clothing	8

CONCLUTION

Tahun 2016 dan 2017, kategori produk dengan jumlah transaksi terendah adalah Bikes. Di tahun berikutnya yaitu 2018 dan 2019, produk dengan penjualan terendah adalah Clothing dan Aksesoris. Perlu dilakukan evaluasi mendalam kenapa produk kategori mendapat penjualan terendah.

BUSSINESS RECOMMENDATION

1. Mengoptimalkan strategi pemasaran yang efektif seperti melakukan cuci gudang untuk produk-produk yang kurang laku di toko.

ACTIONABLE KNOWLEDGE

7. Kota dengan Transaksi Terbesar berdasarkan Kategori Produk

QUERY

```
WITH ranking AS (  
    SELECT  
        dp.category, dg.city, COUNT(t.sales_order_number) AS jumlah_transaksi,  
        SUM(t.totalprice_rupiah - t.totalcost_rupiah) AS profit,  
        row_number() OVER (PARTITION BY dp.category ORDER BY  
COUNT(t.sales_order_number) DESC) AS rank  
    FROM fact_sales t  
    LEFT JOIN dim_product dp ON t.product_id = dp.product_id  
    LEFT JOIN dim_customer dc ON t.customer_id = dc.customer_id  
    LEFT JOIN dim_geography dg ON dc.geography_id = dg.geography_id  
    GROUP BY 1,2  
)  
SELECT * FROM ranking r WHERE r."rank" <= 3
```

OUTPUT

AZ category	AZ city	123 jumlah_transaksi	123 profit	123 rank
Accessories	Cliffside	123	18,630,998	1
Accessories	London	79	13,720,717	2
Accessories	Haney	73	11,051,905	3
Bikes	London	73	957,396,200	1
Bikes	Paris	44	580,800,280	2
Bikes	Warrnambool	34	499,497,480	3
Clothing	Port Hammond	25	5,014,704	1
Clothing	Cliffside	23	3,912,978	2
Clothing	Haney	22	3,806,792	3

CONCLUSION

Kategori produk penyumbang profit terbesar yaitu Bikes, yang masing-masing pembelinya berasal dari London, Paris dan Warrnambool. Namun, untuk jumlah transaksi terbesar yaitu kategori Aksesoris di kota Cliffside sebanyak 123.

BUSSINESS RECOMMENDATION

1. Untuk meningkatkan penjualan “Bikes”, dapat dilakukan promosi di Kota London, Paris dan Warnambool

ACTIONABLE KNOWLEDGE

8. Biaya Produk Terbesar pada Tahun 2017-2019

QUERY

```
WITH ranking AS (  
    SELECT (date_part('year',t.order_date )) AS tahun, dp.category,  
           t.totalcost_rupiah AS total_cost,  
           row_number() OVER (PARTITION BY date_part('year',t.order_date ) ORDER BY  
t.totalcost_rupiah DESC ) AS rank  
    FROM fact_sales t  
    LEFT JOIN dim_product dp ON t.product_id = dp.product_id  
    GROUP BY tahun, 2,3  
)  
SELECT * FROM ranking r  
WHERE r.tahun > 2016 AND r."rank" = 1
```

OUTPUT

123 tahun ▼	A-Z category ▼	123 total_cost ▼	123 rank ▼
2,017	Bikes	30,398,100	1
2,018	Bikes	21,769,300	1
2,019	Bikes	26,573,300	1

CONCLUSION

Kategori *Bikes* secara konsisten menempati peringkat 1 dalam total cost setiap tahun, dengan tren menurun pada 2018 namun kembali naik pada 2019. Hal ini dapat menunjukkan fluktuasi anggaran atau permintaan pasar yang berubah dalam periode tersebut.

BUSSINESS RECOMMENDATION

1. Perusahaan sebaiknya menjaga kualitas, ketersediaan stok, dan strategi pemasaran agar tetap menjadi produk unggulan dan penyumbang utama pendapatan.



Thank You

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